PAPER ID-311828

Subject Code: BCAI053

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM V) THEORY EXAMINATION 2025-26
DATA WAREHOUSING & DATA MINING

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 07 = 14**

Q no.	Question	CO
a.	What are data warehousing components?	1
b.	Differentiate between Database System and Data Warehouse.	1
c.	Explain strategy implemented in Warehousing	1
d.	Explain Client-Server Computing Model.	2
e.	Discuss binning.	2
f.	What is DBSCAN?	1
g.	Describe Aggregation.	2

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

Q no.	Question	CO
a.	Describe the process of building a Data Warehouse. Explain the mapping of the Data Warehouse to a Multiprocessor Architecture.	2
b.	What do you understand by warehouse management and explain the support processes involved in this?	2
c.	Describe Data Integration in data mining and its transformation strategies.	3
d.	Explain Parallel and Distributed Algorithms in data clustering. Differentiate it with Neural Network approach of clustering.	3
e.	Compare and contrast between ROLAP, MOLAP and HOLAP. Explain the advantage and limitation of each approach.	4

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question	CO
a.	How will you explain Multi Dimensional Data Model? Explain Data Cubes, Stars and Snow Flakes.	2
b.	How will you explain Fact Constellation in Data Warehouse modelling? Explain it's advantages and disadvantages.	2

4. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question	CO
a.	Explain the Hardware implemented in Data Warehousing. Discuss the role of Operating Systems for Data Warehousing.	2

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b.	Explain the role of Parallel Processors in Data Warehousing. Define Cluster Systems in view of Data Warehousing.	2

5. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO
a.	Describe Data Cube Aggregation in Data reduction. How will you address Dimensionality reduction in data reduction?	2
b.	What is the role of Decision Tree in data mining? How this help in data reduction?	2

6. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO
a.	What are the Statistical measures in large Databases? Explain Mining Class comparisons in classification.	3
b.	What are Statistical-Based Algorithms? Differentiate it with Distance-Based Algorithms and Decision Tree-Based Algorithms?	3

7. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO
a.	Explain the steps involved in the testing of Data Warehouse. Describe the Tuning Data Warehouse and its specification.	4
b.	Discuss Spatial Mining and Temporal Mining. What are the recent trends in Data Mining and data warehousing that effect the security aspect of the data modeling.	4

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